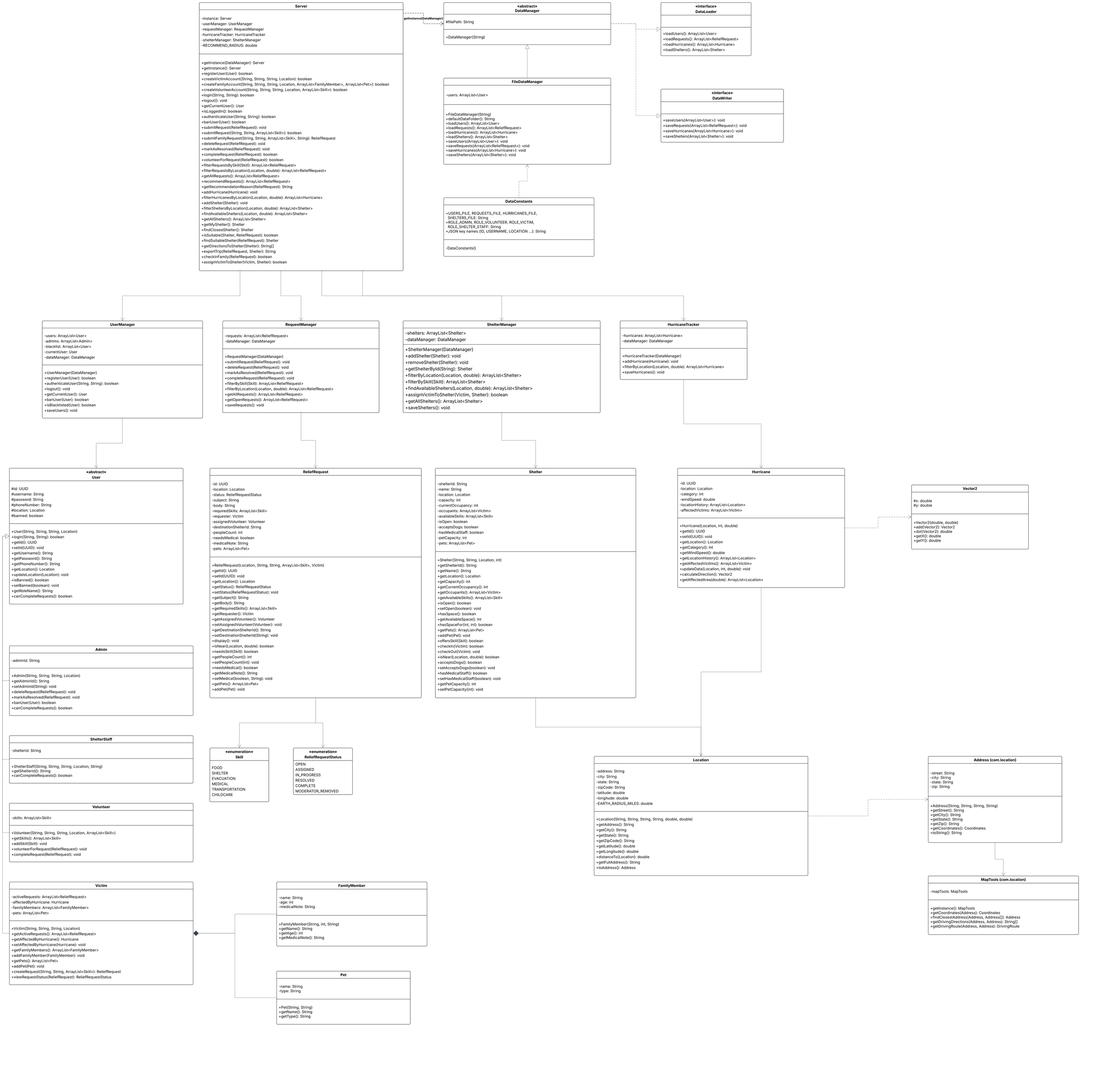

Server
-instance: Server
-userManager: UserManager
-requestManager: RequestManager
-hurricaneTracker: HurricaneTracker
-shelterManager: ShelterManager
-RECOMMEND_RADIUS: double
+getInstance(DataManager): Server
+getInstance(): Server
+registerUser(User): boolean
+createVictimAccount(String, String, String, Location): boolean
+createFamilyAccount(String, String, String, Location, ArrayList<FamilyMember>, ArrayList<Pet>): boolean
+createVolunteerAccount(String, String, String, Location, ArrayList<Skill>): boolean
+login(String, String): boolean
+logout(): void
+getCurrentUser(): User
+isLoggedIn(): boolean
+authenticateUser(String, String): boolean
+banUser(User): boolean
+submitRequest(ReliefRequest): void
+submitRequest(String, String, ArrayList<Skill>): boolean
+submitFamilyRequest(String, String, ArrayList<Skill>, String): ReliefRequest
+deleteRequest(ReliefRequest): void
+markAsResolved(ReliefRequest): void
+completeRequest(ReliefRequest): boolean
+volunteerForRequest(ReliefRequest): boolean
+filterRequestsBySkill(Skill): ArrayList<ReliefRequest>
+filterRequestsByLocation(Location, double): ArrayList<ReliefRequest>
+getAllRequests(): ArrayList<ReliefRequest>
+recommendRequests(): ArrayList<ReliefRequest>
+getRecommendationReason(ReliefRequest): String
+addHurricane(Hurricane): void
+filterHurricanesByLocation(Location, double): ArrayList<Hurricane>
+addShelter(Shelter): void
+filterSheltersByLocation(Location, double): ArrayList<Shelter>
+findAvailableShelters(Location, double): ArrayList<Shelter>
+getAllShelters(): ArrayList<Shelter>
+getMyShelter(): Shelter
+findClosestShelter(): Shelter
+isSuitable(Shelter, ReliefRequest): boolean
+findSuitableShelter(ReliefRequest): Shelter
+getDirectionsToShelter(Shelter): String[]
+exportTrip(ReliefRequest, Shelter): String
+checkInFamily(ReliefRequest): boolean
+assignVictimToShelter(Victim, Shelter): boolean
getInstance(DataManager)
«abstract»
DataManager
#filePath: String
+DataManager(String)
«interface»
DataLoader
+loadUsers(): ArrayList<User>
+loadRequests(): ArrayList<ReliefRequest>
+loadHurricanes(): ArrayList<Hurricane>
+loadShelters(): ArrayList<Shelter>
«interface»
DataWriter
+saveUsers(ArrayList<User>): void
+saveRequests(ArrayList<ReliefRequest>): void
+saveHurricanes(ArrayList<Hurricane>): void
+saveShelters(ArrayList<Shelter>): void
FileDataManager
-users: ArrayList<User>
+FileDataManager(String)
+defaultDataFolder(): String
+loadUsers(): ArrayList<User>
+loadRequests(): ArrayList<ReliefRequest>
+loadHurricanes(): ArrayList<Hurricane>
+loadShelters(): ArrayList<Shelter>
+saveUsers(ArrayList<User>): void
+saveRequests(ArrayList<ReliefRequest>): void
+saveHurricanes(ArrayList<Hurricane>): void
+saveShelters(ArrayList<Shelter>): void
DataConstants
+USERS_FILE, REQUESTS_FILE, HURRICANES_FILE,
SHELTERS_FILE: String
+ROLE_ADMIN, ROLE_VOLUNTEER, ROLE_VICTIM,
ROLE_SHELTER_STAFF: String
+JSON key names (ID, USERNAME, LOCATION ...): String
-DataConstants()
UserManager
-users: ArrayList<User>
-admins: ArrayList<Admin>
-blacklist: ArrayList<User>
-currentUser: User
-dataManager: DataManager
+UserManager(DataManager)
+registerUser(User): boolean
+authenticateUser(String, String): boolean
+logout(): void
+getCurrentUser(): User
+banUser(User): boolean
+isBlacklisted(User): boolean
+saveUsers(): void
RequestManager
-requests: ArrayList<ReliefRequest>
-dataManager: DataManager
+RequestManager(DataManager)
+submitRequest(ReliefRequest): void
+deleteRequest(ReliefRequest): void
+markAsResolved(ReliefRequest): void
+completeRequest(ReliefRequest): void
+filterBySkill(Skill): ArrayList<ReliefRequest>
+filterByLocation(Location, double): ArrayList<ReliefRequest>
+getAllRequests(): ArrayList<ReliefRequest>
+getOpenRequests(): ArrayList<ReliefRequest>
+saveRequests(): void
ShelterManager
-shelters: ArrayList<Shelter>
-dataManager: DataManager
+ShelterManager(DataManager)
+addShelter(Shelter): void
+removeShelter(Shelter): void
+getShelterById(String): Shelter
+filterByLocation(Location, double): ArrayList<Shelter>
+filterBySkill(Skill): ArrayList<Shelter>
+findAvailableShelters(Location, double): ArrayList<Shelter>
+assignVictimToShelter(Victim, Shelter): boolean
+getAllShelters(): ArrayList<Shelter>
+saveShelters(): void
HurricaneTracker
-hurricanes: ArrayList<Hurricane>
-dataManager: DataManager
+HurricaneTracker(DataManager)
+addHurricane(Hurricane): void
+filterByLocation(Location, double): ArrayList<Hurricane>
+saveHurricanes(): void
«abstract»
User
#id: UUID
#username: String
#password: String
#phoneNumber: String
#location: Location
#banned: boolean
+User(String, String, String, Location)
+login(String, String): boolean
+getId(): UUID
+setId(UUID): void
+getUsername(): String
+getPassword(): String
+getPhoneNumber(): String
+getLocation(): Location
+updateLocation(Location): void
+isBanned(): boolean
+setBanned(boolean): void
+getRoleName(): String
+canCompleteRequests(): boolean
ReliefRequest
-id: UUID
-location: Location
-status: ReliefRequestStatus
-subject: String
-body: String
-requiredSkills: ArrayList<Skill>
-requester: Victim
-assignedVolunteer: Volunteer
-destinationShelterId: String
-peopleCount: int
-needsMedical: boolean
-medicalNote: String
-pets: ArrayList<Pet>
+ReliefRequest(Location, String, String, ArrayList<Skill>, Victim)
+getId(): UUID
+setId(UUID): void
+getLocation(): Location
+getStatus(): ReliefRequestStatus
+setStatus(ReliefRequestStatus): void
+getSubject(): String
+getBody(): String
+getRequiredSkills(): ArrayList<Skill>
+getRequester(): Victim
+getAssignedVolunteer(): Volunteer
+setAssignedVolunteer(Volunteer): void
+getDestinationShelterId(): String
+setDestinationShelterId(String): void
+display(): void
+isNear(Location, double): boolean
+needsSkill(Skill): boolean
+getPeopleCount(): int
+setPeopleCount(int): void
+needsMedical(): boolean
+getMedicalNote(): String
+setMedical(boolean, String): void
+getPets(): ArrayList<Pet>
+addPet(Pet): void
Shelter
-shelterId: String
-name: String
-location: Location
-capacity: int
-currentOccupancy: int
-occupants: ArrayList<Victim>
-availableSkills: ArrayList<Skill>
-isOpen: boolean
-acceptsDogs: boolean
-hasMedicalStaff: boolean
-petCapacity: int
-pets: ArrayList<Pet>
+Shelter(String, String, Location, int)
+getShelterId(): String
+getName(): String
+getLocation(): Location
+getCapacity(): int
+getCurrentOccupancy(): int
+getOccupants(): ArrayList<Victim>
+getAvailableSkills(): ArrayList<Skill>
+isOpen(): boolean
+setOpen(boolean): void
+hasSpace(): boolean
+getAvailableSpace(): int
+hasSpaceFor(int, int): boolean
+getPets(): ArrayList<Pet>
+addPet(Pet): void
+offersSkill(Skill): boolean
+checkIn(Victim): boolean
+checkOut(Victim): void
+isNear(Location, double): boolean
+acceptsDogs(): boolean
+setAcceptsDogs(boolean): void
+hasMedicalStaff(): boolean
+setHasMedicalStaff(boolean): void
+getPetCapacity(): int
+setPetCapacity(int): void
Hurricane
-id: UUID
-location: Location
-category: int
-windSpeed: double
-locationHistory: ArrayList<Location>
-affectedVictims: ArrayList<Victim>
+Hurricane(Location, int, double)
+getId(): UUID
+setId(UUID): void
+getLocation(): Location
+getCategory(): int
+getWindSpeed(): double
+getLocationHistory(): ArrayList<Location>
+getAffectedVictims(): ArrayList<Victim>
+updateData(Location, int, double): void
+calculateDirection(): Vector2
+getAffectedArea(double): ArrayList<Location>
Vector2
#x: double
#y: double
+Vector2(double, double)
+add(Vector2): Vector2
+dot(Vector2): double
+getX(): double
+getY(): double
Admin
-adminId: String
+Admin(String, String, String, Location)
+getAdminId(): String
+setAdminId(String): void
+deleteRequest(ReliefRequest): void
+markAsResolved(ReliefRequest): void
+banUser(User): boolean
+canCompleteRequests(): boolean
«enumeration»
Skill
FOOD
SHELTER
EVACUATION
MEDICAL
TRANSPORTATION
CHILDCARE
«enumeration»
ReliefRequestStatus
OPEN
ASSIGNED
IN_PROGRESS
RESOLVED
COMPLETE
MODERATOR_REMOVED
Location
-address: String
-city: String
-state: String
-zipCode: String
-latitude: double
-longitude: double
-EARTH_RADIUS_MILES: double
+Location(String, String, String, String, double, double)
+getAddress(): String
+getCity(): String
+getState(): String
+getZipCode(): String
+getLatitude(): double
+getLongitude(): double
+distanceTo(Location): double
+getFullAddress(): String
+toAddress(): Address
Address (com.location)
-street: String
-city: String
-state: String
-zip: String
+Address(String, String, String, String)
+getStreet(): String
+getCity(): String
+getState(): String
+getZip(): String
+getCoordinates(): Coordinates
+toString(): String
MapTools (com.location)
-mapTools: MapTools
+getInstance(): MapTools
+getCoordinates(Address): Coordinates
+findClosestAddress(Address, Address[]): Address
+getDrivingDirections(Address, Address): String[]
+getDrivingRoute(Address, Address): DrivingRoute
ShelterStaff
-shelterId: String
+ShelterStaff(String, String, String, Location, String)
+getShelterId(): String
+canCompleteRequests(): boolean
Volunteer
-skills: ArrayList<Skill>
+Volunteer(String, String, String, Location, ArrayList<Skill>)
+getSkills(): ArrayList<Skill>
+addSkill(Skill): void
+volunteerForRequest(ReliefRequest): void
+completeRequest(ReliefRequest): void
Victim
-activeRequests: ArrayList<ReliefRequest>
-affectedByHurricane: Hurricane
-familyMembers: ArrayList<FamilyMember>
-pets: ArrayList<Pet>
+Victim(String, String, String, Location)
+getActiveRequests(): ArrayList<ReliefRequest>
+getAffectedByHurricane(): Hurricane
+setAffectedByHurricane(Hurricane): void
+getFamilyMembers(): ArrayList<FamilyMember>
+addFamilyMember(FamilyMember): void
+getPets(): ArrayList<Pet>
+addPet(Pet): void
+createRequest(String, String, ArrayList<Skill>): ReliefRequest
+viewRequestStatus(ReliefRequest): ReliefRequestStatus
FamilyMember
-name: String
-age: int
-medicalNote: String
+FamilyMember(String, int, String)
+getName(): String
+getAge(): int
+getMedicalNote(): String
Pet
-name: String
-type: String
+Pet(String, String)
+getName(): String
+getType(): String